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5604064

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013-14 onwards)

Time : Three hours — Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Form the partial differential equation by eliminating the arbitrary constants from $z = ax + by + a^2 + b^2$.

2. Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$ given that when $x = 0$, $z = e^y$ and $\frac{\partial z}{\partial x} = 1$.

3. Explain boundary value problem.

4. What are the possible solutions of transverse vibrations of a string?

5. What is the one dimensional heat flow equation?

6. Write the Laplace equation.

7. Explain scatter diagram.

8. In two large populations there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations?

9. Mention two uses of chi-square test.

10. Write down the formula for 't' test for difference of means.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. Solve $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$. (11)

Or

12. Solve $(D^3 - 2D^2D')z = 2e^{2x} + 3x^2y$. (11)

Boys :	1	2	3	4	5	6
Marks I test :	23	20	19	21	18	20
Marks II test :	24	19	22	18	20	22
Boys :	7	8	9	10	11	
Marks I test :	18	17	23	16	19	
Marks II test :	20	20	23	20	17	

UNIT II

13. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is in a position given by $y = y_0 \sin^3\left(\frac{\pi x}{l}\right)$. If it is released from rest from this position, find the displacement $y(x, t)$. (11)

Or

14. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is vibration by giving to each of its points a velocity $\lambda x(l - x)$, find the displacement of the string at any distance x from one end at any time t. (11)

UNIT III

15. An isolated rod of length l has its ends A and B maintained at 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained at 0°C, find the temperature at a distance x from A at time t.

Or

16. The bounding diameter of semi-circular plate of radius $a\,cm$ is kept at 0°C and the temperature along the semi-circular boundary is given by

$$u(a, \theta) = \begin{cases} 50\theta, \; when \; 0 < \theta \frac{\pi}{2} \\ 50(\pi - \theta), \; when \frac{\pi}{2} < \theta < \pi \end{cases}$$

Find the steady state temperature function $u(r, \theta)$. (11)

UNIT IV

17. An experiment gave the following values :

v (*ft/min*) :	350	400	500	600
t (*min*)	61	26	7	26

It is know that v and t are connected by the relation $v = at^b$. Find the best possible values of a and b. (11)

Or

18. A die was thrown 9000 times and a throw of 5 or 6 was obtained 3240 times. On the assumption of random throwing, do the data indicate an unbiased die? (11)

UNIT V

19. Fit a normal distribution to the following data weights of 100 students of Delhi University and test the goodness of fit. (11)

Weights (*kg*) :	60-62	63-65	66-68	69-71	72-74
Frequency :	5	18	42	27	8

Or

20. Eleven students were given a test in statistics. They were given a month's further tuition and a second test of equal difficulty was held at the end of it. Do the marks give evidence that the students have benefited by extra coaching? (11)

UNIT IV

17. Fit a second degree parabola to the following data, using method of orthogonal polynomials :

x :	0.5	1.0	1.5	2.0	2.5	3.0
y :	72	110	158	214	290	380

Or

18. A dice is thrown 9,000 times and a throw of 3 or 4 is observed 3,240 times. Show that the dice can be regarded as an unbiased one and find the limits between which the probability of a throw of 3 or 4 lies.

UNIT V

19. Below are given the gain in weights (in lks) of pigs fed on two diets A and B :

Diet A :	25	32	30	34	24	14	32	24	30	31	35	25		
Diet B :	44	34	22	10	47	31	40	30	32	35	18	35	29	22

Test, if the two diets differ significantly as regards their effect on increase in weight.

Or

20. Fit a Poisson distribution to the following data and test the goodness of fit.

x :	0	1	2	3	4	5	6
f :	275	72	30	7	5	2	1

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B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2019.

Fourth Semester

Civil Engineering

MATHEMATICS – IV

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

1. Eliminating arbitrary constant from $z = ax + by + a^2 + b^2$.

2. Eliminating arbitrary function from $z = f\left(\frac{xy}{2}\right)$.

3. What are the assumptions for transverse vibrations of an elastic string?

4. Write the boundary conditions of vibrating string with non-zero initial velocity.

5. Write the one dimensional heat flow equation.

6. Write the two dimensional heat flow equation in polar forms.

7. Write the confidence limits for μ in test of significance for single mean.

8. Fit a straight line to the following data :

x :	1	2	3	4	6	8
y :	2.4	3	3.6	4	5	6

9. Write the probability that person possesses the attribute A_i.

10. Test the hypothesis that $\sigma = 10$, given that $s = 15$ for a random sample of size 50 from the normal population.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. (a) Solve : $p^2 + q^2 = 4$.

(b) Solve : $z = px + qy + \sqrt{1 + p^2 + q^2}$.

Or

12. Solve : $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$.

UNIT II

13. Solve by using method of separation of variables the equation $2x\frac{\partial z}{\partial x} - 3y\frac{\partial z}{\partial y} = 0$.

Or

14. A string is stretched and fastened to two points $x = 0$ and $x = l$ apart. Motion is started by displacing the string into the form $y = k(lx - x^2)$ from which it is released at time $t = 0$. Find the displacement of any point on the string at a distance of x from one end at time t.

UNIT III

15. Solve the equation $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ subject to the boundary conditions $u(0, t) = 0$, $u(l, t) = 0$, $u(x, 0) = x$.

Or

16. Find the steady temperature distribution at points in a rectangular plate with insulated faces the edges of the plate bounded by the lines $x = 0$, $x = a$, $y = 0$ and $y = b$. When three of the edges are kept at temperature zero and the fourth at a fixed temperature a °C.

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UNIT IV

17. Fit a second degree parabola to the following data taking x as the independent variable?

x:	1	2	3	4	5	6	7	8	9
y:	2	6	7	8	10	11	11	10	0

Or

18. In a sample of 500 people in Kerala 280 are tea drinkers and the rest are coffee drinkers. Can we assume that both coffee and tea are equally popular in the state at 5% level of significance?

UNIT V

19. A company keeps records of accidents. During a recent safety review, a random sample of 60 accidents was selected and classified by the day of the week on which they occurred.

Day:	Mon	Tue	Wed	Thu	Fri
No. of accidents	8	12	9	14	17

Test whether there is any evidence that accidents are more likely on some days than others.

Or

20. The following data is collected on two characteristics:

	Smokers	Non-Smokers
Literate	83	57
Illiterate	45	68

Based on this can you say that there is no relation between the habit of smoking and literacy.

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Fourth Semester

Civil Engineering

MATHEMATICS – IV

(From 2013 – 14 onwards)

Time : Three hours — Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Obtain partial differential equation by eliminating the arbitrary constants a and b from $z = a + b(x + y)$.

2. Solve $z = px + qy - 2\sqrt{pq}$.

3. Find the complementary function of $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2 0} = x + y$.

4. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ using variable separable method.

5. Define the term "thermal conductivity".

6. What is the one dimensional heat equation at give the most general solution of it.

7. Write down the normal equations for fitting a curve of the form $y = ax^b$.

8. Explain the term one-tail test.

9. When do we accept the hypothesis H_0 when $t = \dfrac{\bar{x}_1 - \bar{x}_2}{S\sqrt{\dfrac{1}{n} + \dfrac{1}{n_2}}}$ is the case of testing two populations are equal.

10. What is the test statistic χ^2 defined as?

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing One questions from each Unit.

UNIT I

11. Solve $(p^2 + q^2)y = qz$.

Or

12. Solve $(D^3 + D^2D' - DD'^2 - D'^3)z = e^{2x+y} + \cos(x+y)$.

UNIT II

13. Solve $x^2 \dfrac{\partial z}{\partial y} + y^3 \dfrac{\partial z}{\partial x} = 0$ by the method of separation of variables.

Or

14. Show that the solution of the one dimensional wave equation $\dfrac{\partial^2 y}{\partial x^2} = \dfrac{1}{c^2}\left(\dfrac{\partial^2 y}{\partial t^2}\right)$ with the end conditions $y = 0$ at $x = 0$ and $x = l$ can be put in the form $\sum_{m=1}^{\infty} \sin\dfrac{m\pi x}{l}\left(A_m \cos\dfrac{m\pi ct}{l} + B_m \sin\dfrac{m\pi ct}{l}\right)$.

UNIT III

15. A rod of length l is heated so that its ends A and B are at zero temperature. If initially its temperature is given by $u = \dfrac{cx(l-x)}{l^2}$ find the temperature at time t.

Or

16. The initial temperature in a bar with ends $x = 0$ and $x = \pi$ is $u = \sin x$. If the initial surface is insulated and the ends are held at Zero, find the temperature $u(x, t)$.

(b) In a sample of 1000 people in Maharastra 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at 1% level of significance? (5)

Or

18. In a large city A, 20% of a random sample of 900 School children had defective eyesight. In another large city B, 15% of a random sample of 1,600 children had the same defect. Is this difference the two proportions significant? Obtain 95% confidence limits for the difference in the population Proportions. (11)

UNIT V

19. It is believed that the precision of an instrument is no more than 0.16. Write down the null and alternative hypothesis for testing this belief. Carry out the test at 1% level given 11 measurements of the same subject on the instrument : 2.5, 2.3, 2.4, 2.3, 2.5, 2.7, 2.5, 2.6, 2.6, 2.7, 2.5. (11)

Or

20. A Survey of 800 families with four children each revealed the following distributions :

No.of Boys	0	1	2	3	4
No.of Girls	4	3	2	1	0
No. of families	32	178	290	236	64

Is this result consistent with the hypothesis that male and female births are equally probable? (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2018.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Solve $p^2 - 9p + 18 = 0$ where $p = \dfrac{dy}{dx}$.

2. Solve $(D^3 - 3D^2 + 3D - 1)\,y = 0$.

3. Define boundary value problem.

4. Write down the steady state heat flow equation.

5. Define two dimensional heat flow.

6. Write down the Fourier law of heat conduction.

7. Define binomial probability distribution.

8. Under null hypothesis $p_1 = p_2$ then write down the test statistic for the difference of proportions.

9. Define Chi-Square variate with n degrees of freedom.

10. What are the conditions should be satisfied the validity of Chi-Square test?

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. (a) Solve $y = (x-a)p - p^2$. (5)

(b) Solve $x^2(y - px) = yp^2$. (6)

Or

12. Solve $D^3 - 3D^2 + 4D - 2)y = e^x \cos x$. (11)

UNIT II

13. (a) Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2\frac{\partial z}{\partial t} + u$ where $u(x,0) = 6e^{-3x}$. (5)

(b) Solve by the method of separation of variable $4\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 3z$ subject to $z = e^{-5y}$ when $x = 0$. (6)

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Or

14. A taut string of length $2l$ is fastened at both ends. The midpoint of the string is taken to a height of 'b' and then released from rest in that position. Find the displacement of the string at any time. (11)

UNIT III

15. A rectangular plate with insulated surface in 10 cm wide and so long compared to its width that it may be considered infinite in length without introducing an appreciable error. If the temperature along one short edge $y = 0$ is $T(x,0) = 4(10x - x^2)$ degrees, $0 < x < 10$ while the two longer edges $x = 0$ and $x = 10$ as well as the other short edge are kept at 0°c, find the steady-state temperature function $T(x, y)$. (11)

Or

16. Find the temperature $u(x,t)$ in a bar which is perfect insulated laterally whose ends are kept at 0°c and whose initial temperature is in °c given by $f(x) = x(10 - x)$ given that the length of the bar is 10 cm. (11)

UNIT IV

17. (a) A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Show that the S.E of the proportion of bad ones in a sample of this size is 0.015 and deduce that the percentage of bad pineapples in the consignment almost certainly lies between 8.5 and 17.5. (6)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2018.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013 – 14 Onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

1. Obtain partial differential equation by eliminating the arbitrary constants a and b from $z = (x+a)(y+b)$.

2. Explain singular integral.

3. Explain the initial and boundary conditions.

4. Write the boundary and initial conditions for the transverse vibration of a string of length l with fixed ends with initial displacement $f(x)$.

5. Write down the equation of heat flow in two dimensions in polar coordinates.

6. Write all the solutions of Laplace equation in polar coordinates.

7. Write down the normal equations to fit a straight line $y = a + bx$.

8. Give the various steps in testing of a statistical hypothesis.

9. Write the assumptions of t-test for difference of means.

10. Write down the chi square test of independence of (2×2) contingency table.

PART B — $(5 \times 11 = 55$ marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. (a) Form the partial differential equation by eliminating the arbitrary function from the relation $f(xy + z^2, x + y + z) = 0$.

(b) Find the general integral of the equation $\frac{\partial^2 z}{\partial x^2} + 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = x + y$.

Or

0. (a) A random sample of 10 boys had the following I.Q.'s: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

(b) The theory predicts the proportion of beans in the four groups A, B, C and D should be 9 : 3 : 3 : 1. In an experiment among 1600 beans, the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory?

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12. (a) Find the general solution of the partial difference equation

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$$(3z - 4y)p + (4x - 2z)q = 2y - 3x .$$

(b) Solve $(D^2 + 3DD' + 2D'^2)z = \sin(x + 5y)$.

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UNIT II

13. (a) Solve $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ by the method of separation of variables.

(b) State any four assumptions involved in deriving one dimensional wave equation.

Or

14. A string is stretched and fastened to two points l apart. Motion is started by displacing the string into the form $y = k(lx - x^2)$ from which it is released at time $t = 0$. Find the displacement of any point on the string at a distance of x from one end at time t.

UNIT III

15. A rod of length l has its ends A and B kept at 0°C and 100°C respectively, until steady-state conditions prevail. If the temperature at B is reduced suddenly to 0°C and kept so, while that of A is maintained, find the temperature $u(x,t)$ at a distance x from A and at time t.

Or

16. The vertices of a thin square plate are $(0,0)$ $(l,0)$ $(0,l)$ and (l,l). The upper edge of the square is maintained at an arbitrary temperature distribution given by $u(x,l)=f(x)$. The other three edges are maintained at the temperature zero. Find the steady-state temperature at any point within the plate.

UNIT IV

17. Fit a parabola of second degree to the following data :

X:	0	1	2	3	4
Y:	1	1.8	1.3	2.5	6.3

Or

18. (a) In a sample of 1,000 people in Maharash[tra] 540 are rice eaters and the rest are wh[eat] eaters. Can we assume that both rice an[d] wheat are equally popular in this state at 1% level of significance?

(b) The means of two single large samples of 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of standard deviation 2.5 inches? (Test at 5% level of significance).

UNIT V

19. (a) Samples of two types of electric light bulbs were tested for length of life and following data were obtained :

	Type I	Type II
Sample No.	$n_1 = 8$	$n_2 = 7$
Sample Means	$\bar{x}_1 = 1234\ hrs$	$\bar{x}_2 = 1036\ hrs$
Sample S.D.'s	$s_1 = 36\ hrs$	$s_2 = 40\ hrs$

Is the difference in the means sufficient to warrant that type I is superior to type II regarding length of life?

(b) A random sample of 27 pairs of observations from a normal population gave a correlation coefficient of 0.6. Is this significant of correlation in the population?

Or

0.135

18. The average hourly wage of a sample of 150 workers in a plant 'A' was Rs. 2.56 with a standard deviation of Rs. 1.08. The average wage of a sample of 200 workers in plant 'B' was Rs. 2.87 with a standard deviation of Rs. 1.28. Can an applicant safely assume that the hourly wages paid by plant 'B' are higher than those paid by plant 'A'?

UNIT V

19. The following figures show the distribution of digits in numbers chosen at random from a telephone directory

Digits :	0	1	2	3	4	5
Frequency :	1026	1107	997	966	1075	933
Digits :	6	7	8	9	Total	
Frequency :	1107	972	964	853	10,000	

Test whether the digits maybe taken to occur equally frequently in the directory.

Or

20. Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y) :

X :	65	66	67	67	68	69	70	72
Y :	67	68	65	68	72	72	69	71

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B.Tech. DEGREE EXAMINATION, MAY 2017.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Form the partial differential equation by eliminating the arbitrary constants from $Z = ax + by + a^2 + b^2$.

2. Obtain the partial differential equations by eliminating 'f' from $z = xy + f(x^2 + y^2 + z^2)$.

3. What are the possible solutions of one dimensional wave equation?

4. Solve using separation of variables method $yu_x + xu_y = 0$.

5. In steady state conditions derive the solution of one dimensional heat flow equation.

6. Write the two dimensional heat-flow equation in Cartesian coordinate for steady state case.

7. Write down the normal equation in the fining of a straight line.

8. Write down the confidence limits for μ.

9. Define correlation coefficient.

10. What is meant by mean-square contingency?

SECTION B — ($5 \times 11 = 55$ marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. Solve $(mz - ny)p + (nx - lz)q = 1y - mx$.

Or

12. (a) Solve : $(D^2 - 3DD^1 + D^{1^2})z = \sin x \cos y$. (6)

(b) Solve : $(D^2 - D^{1^2} - 3D + 3D^1)z = xy$. (5)

UNIT II

13. Solve by using method of separation of variables the equation $2x\dfrac{\partial z}{\partial x} - 3y\dfrac{\partial z}{\partial y} = 0$.

Or

14. A string is stretched and fastened to two points $x = 0$ and $x = 1$ apart. Motion is started by displacing the string into the form $y = k(1x - x^2)$ from which it is released at time $t = 0$. Find the displacement of any point on the string at a distance of x from one end at time t.

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UNIT III

15. Solve the equation $\dfrac{\partial u}{\partial t} = \alpha^2 \dfrac{\partial^2 u}{\partial x^2}$ subject to the boundary conditions $u(0,t) = 0$, $u(1,t) = 0, u(x,0) = x$.

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Or

16. A square plate is bounded by the lines $x = 0, y = 0, x = 20$ and $y = 20$. Its faces are insulated. The temperature along the upper horizontal edge is given by $u(x,20) = x(20 - x)$ when $0 < x < 20$ while the other three edges are kept at 0°C. Find the steady state temperature in the plate.

UNIT IV

17. A dice is thrown 9,000 times and a throw of 3 or 4 is observed 3,240 times. Show that the dice cannot be regarded as an unbiased one and find the limts between which the probability of a throw of 3 or 4 lies.

Or

5604064

B.Tech. DEGREE EXAMINATION, NOVEMBER 2016.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013–14 onwards)

Time : Three hours Maximum : 75 marks

Statistical tables are permitted.

SECTION A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

All questions carry equal marks.

1. Form the partial differential equation by eliminating the arbitrary constants from $z = (x^2 + a)(y^2 + b)$.

2. Solve $p^2 + q^2 = npq$.

3. State the assumptions made while deriving the equation of Vibration of a String in the simple form.

4. What are the possible solutions of the one dimensional wave equation?

5. What does α^2 represent in the equation $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$?

6. State the two laws of thermodynamics used in the derivation of two dimensional heat flow equations.

7. What is the principle of least square?

8. Explain Type I and Type II errors.

9. What are the assumptions made in the student's t–test?

10. State any two important applications of χ^2 distribution.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full questions from each Unit.

UNIT I

11. (a) Find the singular integral of $z = px + qy + \left(\frac{p}{q} - p\right)$. (5)

(b) Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$. (6)

Or

12. Solve $(D^3 - 7DD'^2 - 6D'^3)z = x^2 y + \sin(x + 2y)$. (11)

20. Fit a Poisson distribution to the following data and test the goodness of fit. (11)

x:	0	1	2	3	4	5	6
f:	275	72	30	7	5	2	1

15|4|18

UNIT II

13. A tightly stretched string has its ends fixed at $x=0$ and $x=l$. At time $t=0$, the initial displacement of the string is given by $f(x)=kx^2(l-x)$ where k is a constant and then released from rest. Find the displacement of any point 'x' of the string at any time t. (11)

Or

14. If a string of length of 20 cm is initially at rest in its equilibrium position and each of its points is given a velocity v such that $v=\begin{cases} x & \text{for } 0<x\le 10 \\ 20-x & \text{for } 10<x\le 20 \end{cases}$. Find the displacement of the string $y(x,t)$. (11)

UNIT III

15. A rod of length l has ends A and B kept at 0°C and 120°C respectively until steady state conditions prevail. If the temperature at B is reduced to 0°C and kept so while that of A is maintained. Find the temperature distribution in the rod. (11)

Or

16. An infinitely long rectangular plate with insulated surface is 20 cm wide. The two long edges and one short edge are kept at 0°C, while the other short edge $x = 0$ is kept at temperature given by $u = \begin{cases} 10y & \text{for } 0 \le y \le 10 \\ 10(20-y) & \text{for } 10 \le y \le 20 \end{cases}$. Find the steady state temperature in the plate. (11)

UNIT IV

17. (a) Fit a parabola to the following data; also estimate y at $x = 6$. (6)

x :	1	2	3	4	5
y :	5	12	26	60	97

(b) A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled, it puts out 3 imperfect articles in batch of 100. Has the machine improved? (5)

Or

18. (a) Fit a curve of the form $y = ab^x$ to the data. (6)

x :	2	3	4	5	6
y :	144	172.8	207.4	248.8	298.5

(b) The following data are got from investigation.

	No. of cases	Mean wages	S.D of the wages
Sample I	400	Rs. 47.4	Rs. 3.1
Sample II	900	Rs. 50.3	Rs. 3.3

Find out whether the two mean wages differ significantly. (5)

UNIT IV

19. 15/4/18 (a) Two random samples of 11 and 9 items show the sample standard deviations of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test the hypothesis that the true variances are equal, against the alternative hypothesis that they are not. (5)

15/4/18 (b) Find if there is any association between extravagance in fathers and extravagance in sons from the following data : (6)

	Extravagant father	Miserly father	
Extravagant son	327	741	1068
Miserly son	545	234	779
	872	975	

Determine the coefficient of association also.

Or

UNIT IV

17. Fit a Parabola $y = ax^2 + bx + c$ in least square sense to the data.

x :	10	12	15	23	20
y :	14	17	23	25	21

Or

18. A machine produced 20 defective units in a sample of 400. After overhauling the machine it produced 10 defective units in a batch of 300. Has the machine improved in production due to overhauling? Test it a 5% L.O.S.

UNIT V

19. A random sample of 10 boys had the following I.Q's 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of population mean I.Q of 100?

Or

20. The following table gives the number of aircraft accidents that occur during the various days of the week. Test whether the accidents are uniformly distributed over the week.

Days :	Mon	Tue	Wed	Thu	Fri	Sat	Total
No. of accidents :	14	18	12	11	15	14	84

B.Tech. DEGREE EXAMINATION, MAY 2016.

Fourth Semester

Civil Engineering

MATHEMATICS — IV

(From 2013-14 batch onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Form the partial differential equation from $z = f(2x - 6y)$.

2. Find the complete integral of $z = px + qy + p^2 - q^2$.

3. Write all possible solutions of one dimensional wave equation.

4. What are the assumptions made in deriving the one dimensional wave equation?

5. A rod of 30 cm long its ends A and B kept at 20°C and 80°C until steady state prevails. Find the steady state temperature distribution.

6. Give the three possible solutions of two dimensional steady state heat flow equation.

7. Define Null hypothesis and Alternative hypothesis.

8. Mention the various steps involved in testing of hypothesis.

9. Write any two applications of t-test.

10. Define χ^2 test of goodness of fit.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the units, choosing ONE full question from each unit.

UNIT I

11. Solve the Partial differential equation $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$.

Or

12. Solve $(D^2 + 4DD' - 5D'^2)z = 3e^{2x-y} + \sin(x-2y)$.

UNIT II

13. A string is stretched and fastened to two points l apart. Motion is started by displacing the string into the form $k(lx - x^2)$ from which it is released at time $t = 0$. Find the displacement of any point on the string at a distance from one end at time 't'.

Or

14. A string is stretched between two fixed points at a distance apart and the points on the string are given initial velocity V, where

$$V = \begin{cases} \frac{c}{l}x, & 0 < x < l \\ \frac{c}{l}(2l - x), & l < x < 2l \end{cases}$$, Find the displacement.

UNIT III

15. A metal bar 10 cm long with insulated sides has its ends A and B kept at 20°C and 40°C respectively until steady state conditions prevail. The temperature at A is then suddenly raised to 50°C and B is lowered to 10°C. Find the subsequent temperature distribution.

Or

16. A rectangular plate with insulated surfaces is 10 cm wide and so long compared to its width that it may be considered infinite in length. The temperature along the short edge $y = 0$ is given by

$$u(x,0) = \begin{cases} 20x, & 0 < x < 5 \\ 20(10 - x), & 5 < x < 10 \end{cases}$$. Other edges

kept at 0°C. Find the steady state temperature distribution.